SENIOR ENGINEERING LEADER

Puspamitra Mishra

Cloud transformation. API platforms. Enterprise AI.

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Project portfolio



PROFILE SUMMARY

Senior engineering leader with 24+ years across healthcare, insurance, banking, and life sciences. Leads global teams of 350+ through cloud migration, API modernization, and enterprise AI adoption. Delivered $6M+ in annualized cloud savings and eliminated approximately $1M a year in legacy licensing.

24+	350+	$48M	$6M+
Years in leading, managing and transforming enterprise IT	Engineers led during various transformation initiatives	Saved through AI-led transformations across industries	Annualized cloud savings from modernization

WORK EXPERIENCE

Sep 2025 - Present
Senior Engineering Leader - API Platforms
CVS Health / Aetna, via TCS
Leads APIC-to-Kong migration and high-throughput API platforms across Azure and GCP; observability sustains 99.99% availability.

Aug 2024 - Aug 2025
Senior Engineering Leader - Cloud Migration
McKesson, via TCS
Directed GCP-to-Azure migration and optimization, saving ~$500K/month ($6M annualized) through a Microsoft-funded program.

Mar 2021 - Jul 2024
Senior Engineering Leader - Modernization & AI
MetLife, via TCS
Led 350+ engineers; Azure microservices eliminated ~$1M/year in ACE licensing and reduced maintenance overhead 30%.

Feb 2017 - Feb 2021
Senior Engineering Manager
Citigroup, via TCS
Migrated core banking to Spring Boot microservices on PCF, AWS, and OpenShift; governed releases and payments.

Dec 2005 - Oct 2017
Solutions Architect / Senior Developer
Citibank, via TCS
Led 35-50 developers; designed mortgage underwriting rules and modernized fees and pricing platforms.

Earlier: Senior Developer at RS Software (2004-05), Satyam Computers (2004), and Algorithm Computers (2002-04).

EDUCATION & PROFESSIONAL CREDENTIALS





B.E., Electronics & Telecommunication · Utkal University, India
Claude Certified Architect - Professional (2026)
TCS Generative AI Executive Level (2025) · AWS Solutions Architect - Associate (2024)
Sun Certified Java Programmer (SCJP - SE 5.0)

SELECTED AI PROJECTS

Life sciences & pharmaceutical intelligence



01 / LIFE SCIENCES

Clinical Research Document Intelligence for Cell & Gene Therapy

Lead AI Solutions Architect

RAG · Multi-agent drafting · Knowledge graph

Architected a GxP-validated copilot combining RAG, multi-agent drafting, and a cell and gene therapy knowledge graph across six programs. Medical writers retain accountability for clinical and regulatory content.

46%	81%	$3.8M
Faster CSR drafting	Less safety narrative effort	Annual benefit

CONCEPTUAL WORKFLOW

Source documents > RAG + CGT knowledge graph > Drafting & checking agents > Medical writer review

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02 / PHARMACEUTICALS

AI Pricing Engine for Generic Pharmaceuticals

AI Solutions Architect & Technical Lead

Predictive models · Gurobi · Salesforce CPQ

Architected segment-aware pricing for ~1,200 generic pharmaceutical SKUs. Demand and bid-win models feed Gurobi optimization, with LLM explanations in Salesforce CPQ and early warnings of competitive price erosion.

+2.6 pts	42%	$15.0M
Gross margin improvement	Large-account bid win rate	Annual benefit

CONCEPTUAL WORKFLOW

SKU + customer signals > Demand & win models > Gurobi optimization > Explanations in CPQ

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Source: supplied Project Highlights Summary; outcomes are reported. Photos are illustrative (Unsplash).

SELECTED AI PROJECTS

Insurance & lending intelligence



03 / HEALTH INSURANCE

End-to-End Agentic Claims Processing Automation

Lead AI Solutions Architect

AWS · Event-driven architecture · Multi-agent AI

Designed an AWS event-driven agent pipeline for 6.5M claims annually, gathering API evidence and checking policy and coding. Exceptions reach evaluators with reasoning; the rules engine remains the system of record and humans review every denial.

6.5M	89%	$14.3M
Claims processed per year	First-pass auto-adjudication	Annual benefit

CONCEPTUAL WORKFLOW

Claim submission > API data gathering > Policy & coding checks > Adjudicate or route

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04 / BANKING & LENDING

AI-Assisted Mortgage Underwriting & Conditions Automation

AI Solutions Architect

Document AI · Verification agents · Encompass

Architected document AI, income and asset verification agents, a guideline copilot, and conditions automation inside Encompass for ~38,000 loans annually. Human underwriters retain credit decisions, with fair-lending testing and SR 11-7 controls.

27 days	+53%	$14.7M
Loan cycle time	Underwriter productivity	Annual benefit

CONCEPTUAL WORKFLOW

Loan documents > Income & asset agents > Guidelines & conditions > Human credit decision

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